



Fig. 4: MRI-compatible pacemaker
(Source: www.medgadget.com)



Fig. 5: Disassembled St. Jude Medical Merlin@home EX1150 transmitter (Source: www.mddonline.com)

the long distance between the pacing centres and the population agglomerates posed various problems.

Initially, Furman and his team transmitted data from a pacemaker through telephone connection using the trans-telephonic monitoring (TTM) method. The TTM device used was a 30-day looping event monitor, about the size of a pager that recorded and captured irregular heartbeats by constantly tracking the heart rhythm. The device provided a rudimentary evaluation of pacing and sensing function and the patient was still required to make in-clinic visits.

The modern, reliable, safe, and secure devices became viable in

the 1970s with the availability of the advanced IEC technology, compact longer lasting batteries, microprocessors with fast speeds, and improved analysis techniques and interpretation algorithms. Biotronik was the first company to market a wireless remote home monitoring system that was FDA-approved. Later, other companies followed. Medtronic introduced CareLink in 2005, Boston Scientific brought out Latitude in 2006, and St. Jude started selling House Call in 2007.

Over the last few years, a number of advanced interpretation algorithms have emerged, improving the scope of analysis and diagnosis, and making the devices more and more complex.

How remote monitoring works

During the traditional follow-up visits to a health centre, a sensing wand is placed by the doctor over

the device to download the data into a dedicated computer provided by the manufacturer. Analysing the data, the doctor learns retroactively the events that have happened, such as a high-energy shock or an abnormal activity of heart or the device. Remote monitoring makes possible the information available wirelessly without the patient visiting the centre.

The remote monitoring systems used earlier were non-wireless and transmitted the data when the patient placed a magnetic sensor over the device. The sensor fed the data into a phone modem to transmit it to the clinic or a remote monitoring service.

The present-day remote monitoring systems are wireless and automated. The data from an implanted device, which is compatible with the mostly proprietary remote monitoring system, is downloaded and shared automatically with a home monitor, located within three metres of the patient, using radio transmissions.

Some systems use a smart phone instead of the home monitoring device. The monitor dials up a phone based internet connection and transmits the data to a central server. The healthcare team accesses the data on their computer or a mobile device using a password-protected secure link. The system can transmit data at specific dates and times and/or can be set to transmit when triggered by the programmed alerts, such as the settings going out of range, delivered shocks, or abnormal heart rhythms.

The transmitted data includes such information as heart rhythm changes, device malfunctioning, fluid retention in the lungs, etc. The receiving devices are programmed to look for the specific alerts triggered by events and notify the information automatically to the doctor via a text message, fax, e-mail, or a website message. The information also gets downloaded into the patient's electronic medical record (EMR).

Remote monitoring provides the healthcare team an easy access to information, enabling them to manage the patient's heart condition and the function of the implanted device in a better and quicker way. The quality of life of patients is improved with reduced hospitalisation and curtailed visits to the clinics, and they also get a sense of security and peace of mind.

Electromagnetic and RF interference

Medical Implants Communication Services (MICS) frequency band,